

1. (Fall 2024, #1) Compute the following limits. No credit will be earned if L'Hopital's rule is used. You are graded on your justification, not your final answer. If a theorem is required, you must explicitly verify the hypothesis of the theorem.

(a) $\lim_{x \rightarrow 0} x^2 e^{\cos(\frac{1}{x})}$

Since $-1 \leq \cos(\frac{1}{x}) \leq 1$ for all x , and e^x is increasing,
we have $e^{-1} \leq e^{\cos(\frac{1}{x})} \leq e$ for all x . So

$$x^2 e^{-1} \leq x^2 e^{\cos(\frac{1}{x})} \leq e x^2$$

$$\rightarrow 0 = \lim_{x \rightarrow 0} x^2 e^{-1} \leq \lim_{x \rightarrow 0} x^2 e^{\cos(\frac{1}{x})} \leq \lim_{x \rightarrow 0} e x^2 = 0$$

$$\rightarrow \boxed{\lim_{x \rightarrow 0} x^2 e^{\cos(\frac{1}{x})} = 0} \quad \text{via the Squeeze Theorem.}$$

(b) $\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 - 5}}{3x + 4}$

$$= \lim_{x \rightarrow \infty} \frac{\sqrt{x^2 - 5}}{-3x + 4} = \lim_{x \rightarrow \infty} \frac{\sqrt{1 - \frac{5}{x^2}}}{-3 + \frac{4}{x}} = \boxed{-\frac{1}{3}}$$

2. (Spring 2024, #1) Compute the following limits. No credit will be earned if L'Hopital's rule is used.

(a)

$$\lim_{x \rightarrow \infty} \frac{\sin(x) + 3 \cos(x)}{x^2}$$

(Hint: use the squeeze theorem.)

$$-4 \leq \sin(x) + 3 \cos(x) \leq 4 \quad \text{for all } x$$

$$\frac{-4}{x^2} \leq \frac{\sin(x) + 3 \cos(x)}{x^2} \leq \frac{4}{x^2} \quad \text{for all } x$$

$$\rightarrow 0 = \lim_{x \rightarrow \infty} \frac{-4}{x^2} \leq \lim_{x \rightarrow \infty} \frac{\sin(x) + 3 \cos(x)}{x^2} \leq \lim_{x \rightarrow \infty} \frac{4}{x^2} = 0$$

$$\rightarrow \boxed{\lim_{x \rightarrow \infty} \frac{\sin(x) + 3 \cos(x)}{x^2} = 0} \quad \text{by the Squeeze Theorem.}$$

(b)

$$\lim_{x \rightarrow 0} \left(\frac{1}{x\sqrt{x+1}} - \frac{1}{x} \right)$$

(Hint: combine the function into one fraction and then consider the conjugate.)

$$\frac{1}{x\sqrt{x+1}} - \frac{\sqrt{x+1}}{x\sqrt{x+1}} = \frac{1 - \sqrt{x+1}}{x\sqrt{x+1}} \cdot \frac{1 + \sqrt{x+1}}{1 + \sqrt{x+1}} = \frac{1 - (x+1)}{x\sqrt{x+1}(1 + \sqrt{x+1})} = \frac{-1}{\sqrt{x+1}(1 + \sqrt{x+1})}$$

$$\lim_{x \rightarrow 0} \frac{-1}{\sqrt{x+1}(1 + \sqrt{x+1})} = \lim_{x \rightarrow 0} \frac{-1}{\sqrt{0+1} \cdot (1 + \sqrt{0+1})} = \boxed{\frac{-1}{2}}$$

3. (Spring 2024, #2) Find the values of a and b that make $f(x)$ continuous everywhere. Show all work in finding the values of a and b .

$$f(x) = \begin{cases} \frac{x^2 - 4}{x - 2} & x < 2 \\ ax^2 - bx + 3 & 2 \leq x < 3 \\ 2x - a - b & x \geq 3 \end{cases}$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2^-} \frac{(x+2)(x-2)}{x-2} = \lim_{x \rightarrow 2^-} x+2 = 4$$

$$> 4a - 2b + 3 = 4$$

$$\lim_{x \rightarrow 2^+} f(x) = f(2) = a \cdot 2^2 - b \cdot 2 + 3 = 4a - 2b + 3$$

$$\text{or} \\ 4a - 2b = 1$$

$$\lim_{x \rightarrow 3^-} f(x) = a \cdot 3^2 - b \cdot 3 + 3 = 9a - 3b + 3$$

$$> 9a - 3b + 3 = 6 - a - b$$

$$\lim_{x \rightarrow 3^+} f(x) = f(3) = 2 \cdot 3 - a - b = 6 - a - b$$

or

$$10a - 2b = 3$$

$$\begin{array}{r} 10a - 2b = 3 \\ - \quad 4a - 2b = 1 \\ \hline \end{array}$$

$$6a = 2 \rightarrow a = \frac{1}{3} \rightarrow 4 \cdot \left(\frac{1}{3}\right) - 2b = 1$$

$$\rightarrow b = \frac{1}{6}$$

$$\boxed{\begin{array}{l} a = \frac{1}{3} \\ b = \frac{1}{6} \end{array}}$$

4. (Fall 2023, #1) Compute the following limits if they exist (they may be finite or infinite). You are not allowed to use L'Hôpital's rule, if you know what that is. You must justify your answers.

(a) $\lim_{x \rightarrow -1} \frac{x^2 - 3x - 4}{x + 1}$

$$= \lim_{x \rightarrow -1} \frac{(x+1)(x-4)}{x+1} = \lim_{x \rightarrow -1} x-4 = -1-4 = \boxed{-5}$$

(b) $\lim_{x \rightarrow 0} \frac{\sin(x^2)}{x \tan(3x)}$

$$= \lim_{x \rightarrow 0} \frac{\sin(x^2)}{x^2} \cdot \frac{3x}{\sin(3x)} \cdot \frac{\cos(3x)}{3}$$

$$= 1 \cdot 1 \cdot \frac{1}{3} = \boxed{\frac{1}{3}}$$

(c) $\lim_{x \rightarrow 2} \frac{\sqrt{x^2+5}-3}{x-2} \cdot \frac{\sqrt{x^2+5}+3}{\sqrt{x^2+5}+3}$

$$= \lim_{x \rightarrow 2} \frac{x^2+5-9}{(x-2)(\sqrt{x^2+5}+3)} = \lim_{x \rightarrow 2} \frac{(x+2)(x-2)}{(x-2)(\sqrt{x^2+5}+3)} = \frac{2+2}{\sqrt{2^2+5}+3} = \boxed{\frac{2}{3}}$$

5. (Fall 2022, #1) Find each limit, including $\pm\infty$ or say it does not exist.

$$(a) \lim_{x \rightarrow 2^-} \frac{|3x^2 - 3x - 18|}{5x^2 - 5x - 30}$$

$$= \lim_{x \rightarrow 2^-} \frac{3 \cdot |x-3| \cdot |x+2|}{5 \cdot (x+2)(x-3)} = \frac{3 \cdot |2-3| \cdot |2+2|}{5 \cdot (2+2) \cdot (2-3)} = \boxed{-\frac{3}{5}}$$

$$(b) \lim_{x \rightarrow 0} \frac{\sin(3x) \ln(x + \pi/4)}{8x^2 + 2\pi x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin(3x)}{3x} \cdot \frac{3 \ln(x + \frac{\pi}{4})}{8x + 2\pi}$$

$$= 1 \cdot \frac{3 \ln(0 + \frac{\pi}{4})}{8 \cdot 0 + 2\pi} = \boxed{\frac{3 \ln(\frac{\pi}{4})}{2\pi}}$$

$$(c) \lim_{x \rightarrow \infty} \left[\frac{\sin(x)}{x} \right]^2$$

$$0 \leq \sin^2(x) \leq 1 \quad \text{for all } x$$

$$0 \leq \frac{\sin^2(x)}{x^2} \leq \frac{1}{x^2} \quad \text{for all } x > 0$$

$$\rightarrow 0 \leq \lim_{x \rightarrow \infty} \frac{\sin^2(x)}{x^2} \leq \lim_{x \rightarrow \infty} \frac{1}{x^2} = 0$$

$$\rightarrow \boxed{\lim_{x \rightarrow \infty} \left[\frac{\sin(x)}{x} \right]^2 = 0} \quad \text{via the Squeeze Theorem.}$$